
Area and Perimeter — Finding and Fixing Area/Perimeter Mix-Ups

Two questions sound almost the same but ask for different things.

- **Perimeter** is the distance all the way *around* the outside. You **add** the sides. The answer is a length: cm or m.
- **Area** is how many unit squares *cover* the inside. For a rectangle you **multiply** the long side by the short side. The answer is a number of squares: cm^2 or m^2 .
- Before you compute, underline the words in the question that tell you which one it wants: *around, fence, ribbon, edging* means perimeter; *cover, fill, tiles, space inside* means area.

1. Perimeter. A name card is 8 cm long and 3 cm wide. Ali glues ribbon all the way around the outside edge of the card. How many cm of ribbon does she use?

Answer: _____ cm

2. Area. The same name card is 8 cm long and 3 cm wide. Ali covers the whole card with one-centimetre squares. How many squares does it take?

Answer: _____ cm^2

3. Find and fix. Milo needs the *perimeter* of a photo that is 6 cm long and 4 cm wide. He wrote $6 \times 4 = 24$ and said the perimeter is 24 cm.

Which measurement did Milo actually find? Cross out his work, then find the perimeter the right way.

Answer: _____ cm

4. Find and fix. Sara needs the *area* of a rug that is 9 m long and 2 m wide. She wrote $9 + 2 + 9 + 2 = 22$ and said the area is 22 square m.

Which measurement did Sara actually find? Cross out her work, then find the area the right way.

Answer: _____ m²

5. Fence around a square. A square garden has four sides that are all the same length. Each side is 7 m long. A fence goes all the way around the garden. How many m of fence are needed?

Answer: _____ m

6. Which one does the question want? Mr. Ruiz has a rectangular play mat that is 10 m long and 6 m wide. He wants soft edging along every outside edge of the mat.

First circle the word that names what the question wants: **perimeter** / **area**. Then find how many m of edging he needs.

Answer: _____ m

7. Same fence, different space. Ana's garden is a square with sides of 5 m. Ben's garden is a rectangle 8 m long and 2 m wide. Both gardens need exactly the same length of fence around them — check that for yourself first.

How many more square m of growing space does Ana's garden have than Ben's?

Answer: _____ m²

8. Show what you know. On the grid below:

- (a) Draw two *different* rectangles whose perimeters are the same but whose areas are different. Use whole grid squares.
- (b) Under each rectangle write its perimeter and its area, with the correct kind of unit for each.
- (c) Write one sentence a classmate could use to decide whether a question is asking for area or for perimeter.

